Chenxi Feng

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EDUCATION

Cornell University — Ithaca, United States

Ph.D. in Atmospheric Science — *09/2024 - Present*

Nanjing University — Nanjing, China

B.S. in Atmospheric Science — *09/2020 - 06/2024*

Thesis: Improvements and Applications of Methane Retrieval Approach Using Imaging Spectrometer

RESEARCH INTERESTS

Land-based carbon sequestration

Solar geoengineering

Climatological sensitivity of land-based carbon dioxide removal

AI-based data-model fusion

Methane point source emission retrieval

PUBLICATIONS

(*, co-first author)

2026 **Feng, C.**, Zhang, S., Qin, S., Xu, X., Tao, F., Luo, Z., ... & Luo, Y. Global convergence in temperature response of microbial respiratory soil organic carbon loss. *In prep.*

2026 Li, F.*, **Feng, C.***, Juan, J. R., Guanter, L., Chen, H., Sun, S., ... & Zhang, Y. An optimized methane retrieval approach based on morphological fusion for mapping methane point emissions from spaceborne imaging spectroscopy. *Remote Sensing of Environment*. 10.1016/j.rse.2026.115718

2026 Gang, C., Wei, N., **Feng, C.**, Xu, H., Liu, H., Tao, F., ... & Luo, Y. Net primary productivity orchestrates uncertainty sources driving global soil organic carbon under land use change. *Communications Earth & Environment*. 10.1038/s43247-026-03312-6

2026 Li, F., Lin, K., Yan, Y., Bai, S., Huang, Q., **Feng, C.**, ... & Zhang, Y. Comparing the performance of different hyperspectral satellite imaging spectroscopy in mapping methane point-source emissions. *Remote Sensing of Environment*. 10.1016/j.rse.2025.115224

2024 **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L. Aerosol-Calibrated matched filter method for retrievals of methane point source emissions over the Los Angeles Basin. *Earth and Space Science*. 10.1029/2024EA003519

2024 Li, F., Bai, S., Lin, K., **Feng, C.**, Sun, S., Zhao, S., ... & Zhang, Y. Satellite-based surveys reveal substantial methane point-source emissions in major oil & gas basins of North America during 2022-2023. *Journal of Geophysical Research: Atmospheres*. 10.1029/2024JD040870

PRESENTATION

2026 **Feng, C.**, Zhang, S., Qin, S., Xu, X., Tao, F., Luo, Z., Yang, Y., Luo, Y. Substrate-induced attenuation of thermal accelerating in soil organic carbon mineralization, 2026 Ecological Society of America Meeting, 26-31 July 2026, Salt Lake City, UT, Poster

2024 **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L. Assessing aerosol-induced bias in methane point source emission retrievals in the LA basin, 2023 American Geophysical Union Fall Meeting, 11-15 December 2023, San Francisco, CA, Poster.

RESEARCH BACKGROUNDS

Cornell University Ithaca, NY, U.S.

PhD student (advisor: Prof. Daniele Visioni)

Efficacy solar geoengineering in mitigating compound hot-dry climate risk on terrestrial carbon sequestration *09/2026 – Present*

- Conducted state-of-art Earth system modeling on terrestrial carbon sequestration response to stratospheric aerosol injection
- Validated climate model simulations with observations from global dimming era

Cornell University Ithaca, NY, U.S.

PhD student (advisor: Prof. Yiqi Luo)

Global convergence in responses of soil organic carbon loss to warming *09/2024 – 09/2026*

- Performed Bayesian inversion analysis on soil incubation dataset
- Constrained a Community Land Model Version 5 (CLM5)-type soil organic carbon (SOC) cycle model
- Identified convergent underlying kinetics controlling divergent responses of microbial mineralization-induced SOC loss to warming

California Institute of Technology Pasadena, CA, U.S.

Research Intern (advisor: Prof. Yuk L. Yung) *06/2023 - 01/2024*

Assessing and eliminating impacts of aerosol on methane point source emission retrieval

- Simulated radiance spectra with the 2S-ESS RT model to study aerosol-induced bias on methane concentration estimates
- Developed an aerosol-calibrated matched filter method for methane retrieval and optimized retrieval accuracy
- Applied the algorithm to methane point sources in Los Angeles using remote sensing data, indicating emission underestimation and proposing improvements

Nanjing University Nanjing, China

Research Assistant (advisor: Prof. Yongguang Zhang) *02/2022 - 07/2024*

Atmospheric methane inversion and point source quantification using hyperspectral observation

- Enhanced methane retrieval accuracy by combining results from weak and strong absorption bands
- Conducted end-to-end simulations to validate proposed methods using ground measured data
- Processed remote sensing data from Chinese satellites to retrieve methane point source emissions
- Analyzed methane plumes in key regions, detecting leaks from oil and gas industries in the US and Middle East

Nanjing University Nanjing, China

Research Assistant (advisor: Prof. Huilin Chen) *12/2022 - 12/2023*

In-situ detection of atmospheric methane and carbon dioxide using unmanned aerial vehicle

- Designed an unmanned aircraft methane and carbon dioxide monitoring system using Axetris AG and K96 sensors
- Conducted lab experiments to assess sensor performance, including sensitivity, accuracy, and response time to gases
- Assisted in planning and preparing field experiments over methane emitters in Nanjing City
- Analyzed methane plumes using the Gaussian model and estimated methane emission flux rate

ACHIEVEMENTS AND AWARDS

Outstanding Thesis Award (The First Prize), Nanjing University	*2024*
Chenxue Scholarship for Overseas Study, Nanjing University	*2024*
Scholarship of Academic Excellence (The First Prize, Top 5%), Nanjing University	*2023*
Zhenggang Fellowship for Oversea Studies of Nanjing University (Top 5%), Nanjing University	*2022*
Scholarship of Academic Excellence (The Second Prize, Top 10%), Nanjing University	*2022*
Outstanding Student for Theory and Methods of Land Surface Remote Sensing Inversion Summer School 2022, Beijing Normal University	*2022*
Scholarship of Academic Excellence (The Second Prize, Top 10%), Nanjing University	*2021*

PROGRAMMING SKILLS

Python, MATLAB, R, FORTRAN